From Words to Worlds: Text-to-Image Generation with Transformers and Diffusion

IE 7615 · Northeastern University · Fall 2026 · Group 1 · Milestone 1 Proposal
GitHub: https://github.com/anshuman-dataverse/text-to-image-generation

1. Objective

Build a text-to-image pipeline in which a Transformer text encoder conditions a pre-trained latent diffusion model, then measure how the conditioning choices (classifier-free guidance scale, noise scheduler, number of sampling steps, text encoder) trade off image fidelity, diversity and prompt alignment. We evaluate quantitatively (FID, Inception Score, CLIP score) and qualitatively, and discuss bias and misuse risks.

2. Dataset

MS COCO Captions 2017, validation split. We use 5,000 natural photos of everyday multi-object scenes (80 categories). They are the standard reference set for text-to-image FID.

- **Working subset: 5,000 image–caption pairs**, one caption per image picked at random with seed 42 (`results/pairs_manifest.csv` lists image ID, caption ID and caption). The full pool of 25,000 cleaned captions (about 5 per image) is kept for the embedding checks.
- **Cleaning:** whitespace and punctuation normalized, captions under 3 words and duplicates removed (25,014 → 25,000), and annotator placeholder captions excluded from the pairs. Verified: 0 missing images, 0 empty captions, 0 duplicate IDs.
- **Preprocessing:** shorter side resized to 512, center-cropped to 512×512 and normalized to [–1, 1]. Captions use CLIP BPE tokens padded to 77 (max 58, none truncated).

3. Model architecture

Component	Model	Role
Text encoder	CLIP ViT-L/14 text transformer (123M)	caption → 77×768 token embeddings
Denoiser	Stable Diffusion v1.5 U-Net (860M)	predicts noise, cross-attending to the text embeddings
Autoencoder	SD v1.5 VAE	512×512 image ↔ 64×64×4 latent
Sampling	DDIM, DDPM, PNDM, Euler, Euler-A, DPM-Solver++ + classifier-free guidance	reverse diffusion

Pipeline: *text prompt → CLIP tokenizer → text encoder → embeddings → U-Net + scheduler → VAE decoder → image.*

4. Methodology

1. **Milestone 1 (done):** prepare data, set up the environment (PyTorch, 🤗 Diffusers, Colab T4), build the text→embedding pipeline and check it. Same-image caption similarity is 0.656 vs. 0.234 for different images, and text→image Recall@1/5/10 is 54.9 / 81.1 / 89.7% among 1,000 images. We ran 5 seeded test prompts, and our own embeddings drive the U-Net identically to SD's built-in encoder.
2. **Milestone 2:** baseline generation over COCO captions; sweep guidance scale $w \in \{1, 3, 5, 7.5, 10, 15\}$ and schedulers × steps {10, 25, 50}.
3. **Milestone 3:** FID and IS (torchmetrics) on 1–2k generated images against COCO references, plus CLIP score, sensitivity plots and a text-encoder comparison (CLIP vs. BERT).
4. **Milestone 4:** final analysis of fidelity vs. diversity, failure modes (object omission, attribute leakage) and ethics.

5. Team and responsibilities

Member	Responsibilities
Anshuman Singh (002590892)	model integration, diffusion image generation, repository management
Rohan Prakash Krishna Prakash (002317798)	dataset collection, cleaning, preprocessing and validation
Sree Ramya Pasala (002301508)	CLIP embedding pipeline, evaluation, visualizations, technical documentation

6. Expected deliverables

Reproducible code and Colab notebook; cleaned 5,000-pair COCO subset manifest; generated image galleries for each configuration; FID / IS / CLIP-score tables and sensitivity plots; milestone reports, final report, slides and demo video (Dec 13).

Project Proposal & Setup – From Words to Worlds (Milestone 1)

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Group 1

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PDF version: Milestone1_Report.pdf · One-page proposal: Milestone1_Proposal.pdf

Repository

GitHub: https://github.com/anshuman-dataverse/text-to-image-generation

Code, results and figures are in the milestone1/ folder of the repository. The whole milestone runs on Google Colab from milestone1_colab.ipynb.

1. Summary

We are building a text-to-image pipeline in which a Transformer text encoder (**CLIP ViT-L/14**) conditions a pre-trained latent diffusion model (**Stable Diffusion v1.5**). We will then measure how the conditioning choices (guidance scale, noise scheduler, number of sampling steps, text encoder) trade off image fidelity, diversity and prompt alignment. Milestone 1 delivers the dataset, the text→embedding pipeline, embedding sanity checks and the first 5 prompt→image samples.

The pipeline ran end to end on a Colab Tesla T4. All **5,000 COCO val2017 images** and 25,000 of the 25,014 captions passed cleaning (14 were exact duplicates). The working subset is **5,000 image–caption pairs**, one caption per image, and no caption exceeds CLIP's 77-token window (max 58). The CLIP embeddings behave as conditioning should. Captions of the same image are far closer than captions of different images (cosine 0.656 vs. 0.234, pairwise AUC 0.985). CLIP retrieves the right image from its caption among 1,000 candidates **54.9%** of the time at rank 1 and **89.7%** within the top 10 (chance 0.1% / 1%). Stable Diffusion produced 5 plausible 512×512 samples at 6.7 s/image. They already show the classic failure modes we will study: a dropped object (the frisbee) and attribute leakage (the "blue car" became a blue-fronted bus). Feeding our own embeddings to the U-Net first exposed a subtle bug: masking CLIP's padding tokens changes 60+ of the 77 conditioning positions and degrades the images. After the fix, our embeddings match SD's own encoding bit for bit and produce pixel-identical images (Section 6).

2. Objective

Generate images from text with a Transformer encoder and a diffusion model, then evaluate them quantitatively (FID, Inception Score, CLIP score) and qualitatively (prompt–image alignment), and analyze

fidelity vs. diversity and the ethical risks. Milestone 1 covers the dataset choice and preprocessing, the environment, the text→embedding pipeline and the baseline test runs.

3. Dataset

3.1 Choice: COCO Captions 2017 (val)

Property	Value
Images	5,000 natural photos, 80 object categories, multi-object everyday scenes
Captions	~5 human-written captions per image (25,014 total)
Working subset	**5,000 image–caption pairs**, one caption per image (`results/pairs_manifest.csv`)
Why	standard reference set for text-to-image FID (comparable to published numbers); compositional scenes test alignment; 5k images fit the 1k–5k scope without the 18 GB train split

Alternatives considered: Flickr30k (people-centric, rarely used for FID) and a LAION-400M subset (noisy web alt-text, and broken links make it hard to reproduce).

3.2 Cleaning and preprocessing

Step	Detail
Captions	whitespace normalization, capitalized, single trailing period, exact duplicates per image removed, captions under 3 words dropped
Subset	seeded random sample (seed 42) of *N* images (default all 5,000)
Pairs	one caption per image, random pick with seed 42, skipping annotator placeholders such as “I am unable to see an image above.” (14 in the pool)
Images	RGB, shorter side resized to 512 (bicubic), center-crop 512×512, which is SD v1.5’s native resolution
Tensors	`ToTensor` + `Normalize(0.5, 0.5)` → [-1, 1], the VAE input range
Tokens	CLIP BPE tokenizer, padded/truncated to 77 tokens

Statistic (results/dataset_stats.json)	Value
Images / captions after cleaning	5,000 / 25,000 (from 25,014; 14 exact duplicates removed)
Image–caption pairs (results/pairs_manifest.csv)	5,000
Captions per image	5.0
Words per caption: mean (min–max)	10.44 (7–50)
CLIP tokens per caption: mean / max	13.78 / 58
Captions truncated at 77 tokens	0
Image size	512×512

COCO captions are short and literal (about 10 words), so truncation is never an issue. The real constraint is that the prompts describe ordinary scenes rather than artistic styles. That makes COCO a good test of *content* alignment, and a poor test of style control.

Pair subset checks (results/subset_checks.json): 0 missing or unreadable images (all 512×512), 0 empty captions, 0 placeholder captions, 0 duplicate image IDs, 0 duplicate caption IDs. 21 caption texts appear on two or three different images (for example “A man holding a tennis racquet on a tennis court.”). They are genuine captions of different photos and are kept. The 25,000-caption pool is used for the embedding checks in Section 5, and the 5,000 pairs are the generation and evaluation set for Milestones 2–3.

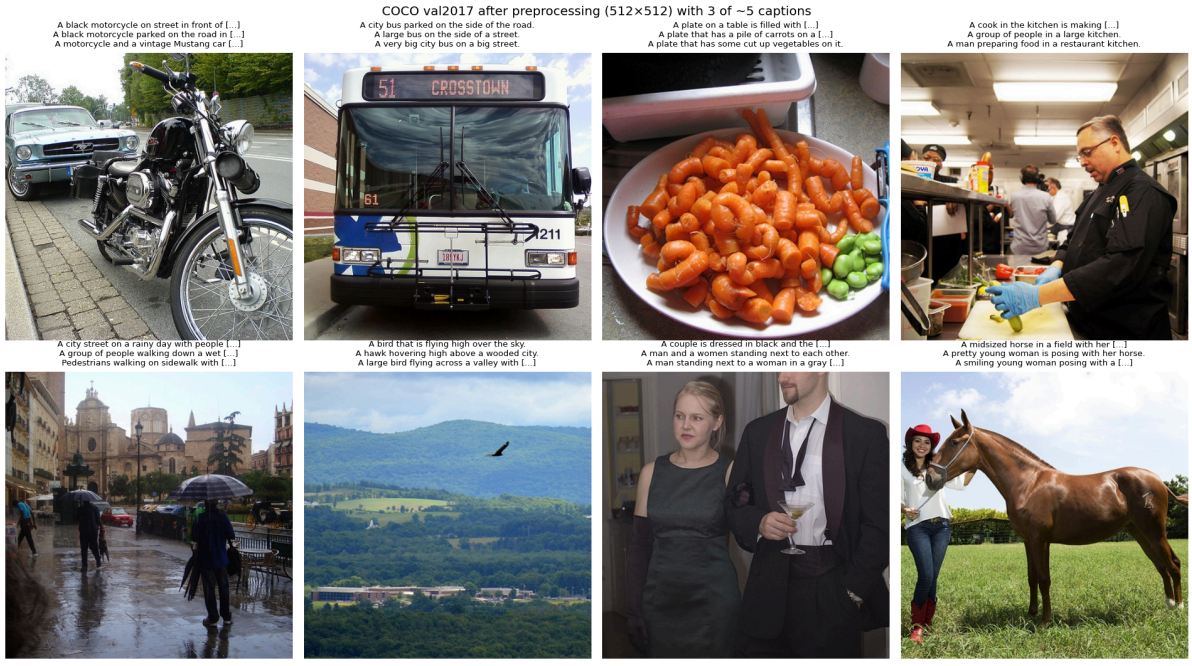


Figure 1. *Example COCO val2017 images after preprocessing (512×512 center crop) with 3 of their ~5 cleaned captions.*

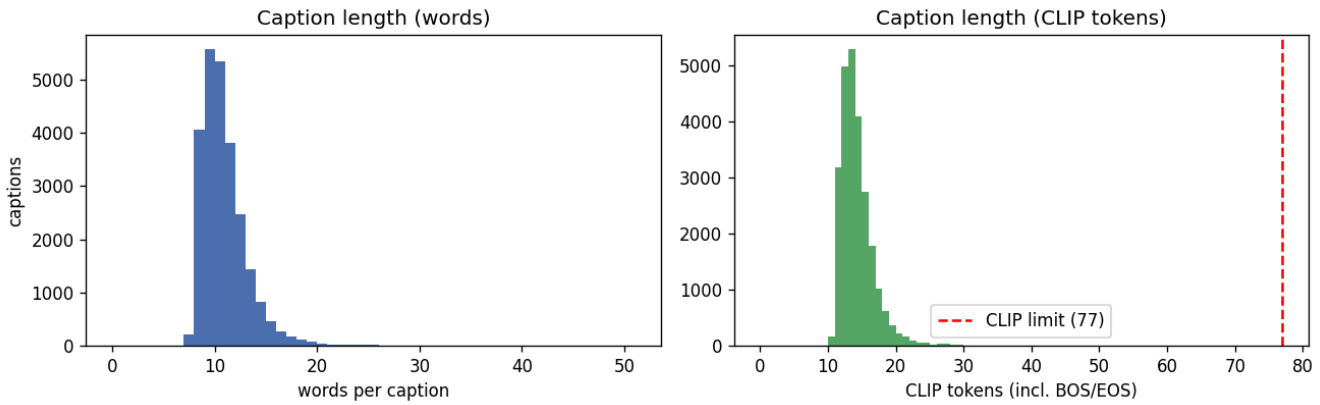


Figure 2. *Caption length in words (left) and CLIP tokens (right). The dashed line is CLIP's 77-token limit; no caption reaches it.*

4. Model

Component	Model	Role
Text encoder	CLIP ViT-L/14 text transformer (123M params, 12 layers)	caption → per-token embeddings (77×768) + pooled embedding (768)
Denoiser	SD v1.5 U-Net (860M params) with cross-attention	predicts noise ε(x_t, t, c), attending to the 77×768 text context
Autoencoder	SD v1.5 VAE	512×512×3 image ↔ 64×64×4 latent
Schedulers	DDIM, DDPM, PNDM, Euler, Euler-A, DPM-Solver++	reverse-diffusion samplers sharing the trained β-schedule
Guidance	classifier-free, $\hat{\varepsilon} = \varepsilon(x_t, \varnothing) + w\cdot(\varepsilon(x_t, c) - \varepsilon(x_t, \varnothing))$	trades alignment for diversity

Our own encoder wrapper (`t2i/embeddings.py`) produces the conditioning tensor, and `t2i/generate.py` passes it to the U-Net as `prompt_embeds`. This lets later milestones swap or modify the text representation (BERT, interpolation, prompt weighting).

Environment. Python with PyTorch (≥ 2.1) and torchvision, 🤗 Diffusers (≥ 0.30) for Stable Diffusion and the schedulers, 🤗 Transformers (≥ 4.44) for CLIP (tokenizer, text encoder and the image tower used for retrieval), and Accelerate and safetensors for model loading (`requirements.txt`). Everything runs on Google Colab with a Tesla T4 (16 GB) in fp16. The notebook installs `requirements.txt`, mounts Google Drive for the COCO data and outputs, and logs the git commit of the code used in every run (`runs/*.log`). Both models are public on the Hugging Face Hub, so no access token is needed.

5. Text → embedding pipeline and sanity checks

1. **Caption similarity:** captions of the same image should be closer than captions of different images.
2. **Text→image retrieval:** CLIP's text and image towers share one space, so each image should be found from its own caption among the candidate images.

Check (results/embedding_checks.json)	Result
Caption cosine similarity, same image (1,000 images, 5,000 captions)	0.656
Caption cosine similarity, different images	0.234
P(same-image pair > different-image pair)	0.985
Text→image Recall@1 / @5 / @10 (1,000 images)	54.9% / 81.1% / 89.7%
Chance @1 / @5 / @10	0.1% / 0.5% / 1.0%
Median rank of the correct image	1

Retrieval is 549× better than chance at rank 1. The misses are informative: for “A living area with a television and a table” all five top images are living rooms with TVs, but not the right one. COCO captions often fit several images equally well, so Recall@1 understates alignment. This matters for Milestone 3, where CLIP score will judge generated images. CLIP rewards matching the caption’s content, not reproducing the reference photo.

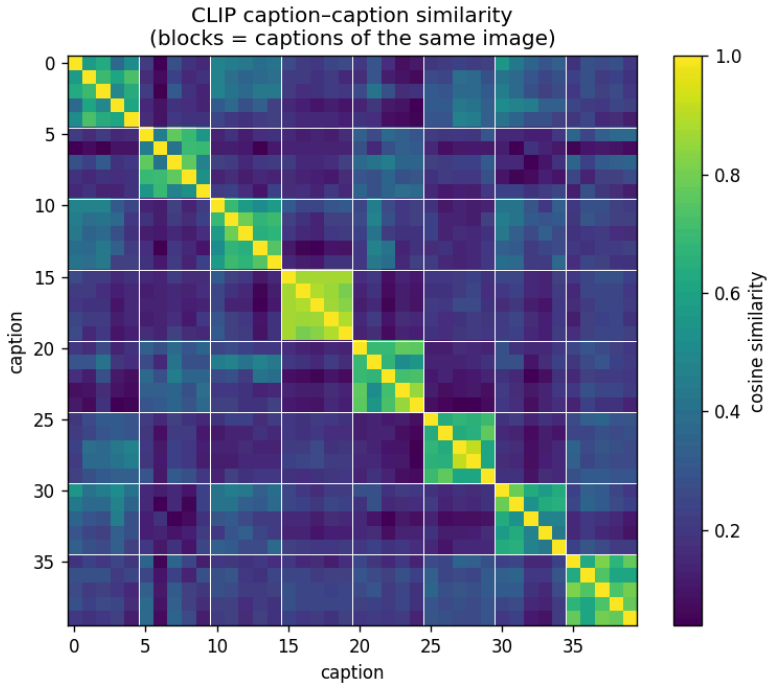


Figure 3. *Cosine similarity between pooled CLIP caption embeddings. Bright diagonal blocks are captions of the same image.*



Figure 4. *CLIP text→image retrieval: top-5 images for a caption, with the ground-truth image outlined in green.*

6. First test runs (5 prompts)

Settings: SD v1.5, DDIM, 30 steps, guidance 7.5, 512×512, seed 0 + i, fp16 on a Tesla T4, **6.7 s per image** (with attention slicing on).

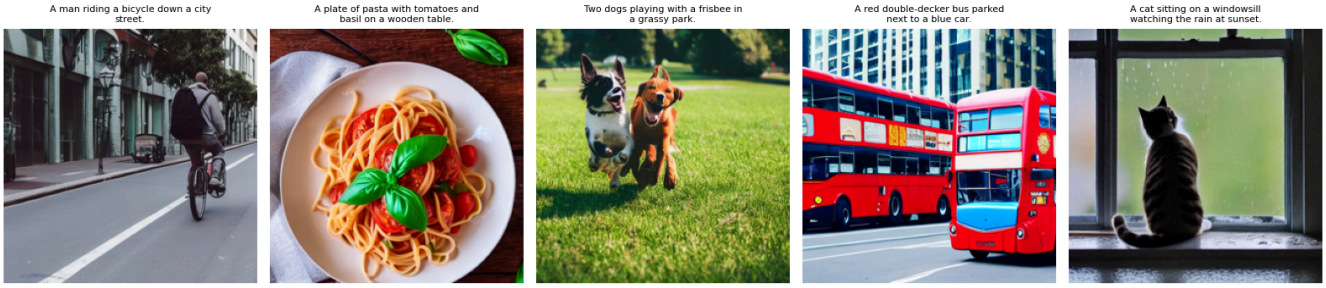


Figure 5. *The five test prompts (SD v1.5, DDIM, 30 steps, w = 7.5, seeds 0–4).*

Prompt	Correct	Wrong / missing
A man riding a bicycle down a city street	rider, bicycle, street, lane markings, buildings	n/a; photorealistic and coherent
A plate of pasta with tomatoes and basil on a wooden table	all four objects, plausible food styling	n/a
Two dogs playing with a frisbee in a grassy park	**count is right** (two dogs), park, playful pose	**frisbee missing** (object omission)
A red double-decker bus parked next to a blue car	red double-decker bus	**no blue car**: the blue leaked onto a second bus's front (attribute leakage)
A cat sitting on a windowsill watching the rain at sunset	cat, windowsill, raindrops on the glass	"sunset" is ignored (grey daylight); the third concept in the prompt is dropped

The model handles single-subject scenes well and degrades as prompts list more objects and attributes. This is the known weakness of CLIP-conditioned diffusion with compositional prompts, and Milestone 2 will test whether higher guidance helps.

Conditioning check: our embeddings → U-Net

We encode the 5 prompts with our own `TextEncoder` and pass the 77×768 tensor to the U-Net as `prompt_embeds`, using the same seeds as above. If our encoding matches SD's, the images must be identical.

First run (bug): the images differed by 47–78 / 255 on average, and three of them broke down completely (below). Our encoder passed CLIP an `attention_mask`, but SD's pipeline does not. Because CLIP's text transformer is causal, the real tokens and the pooled EOS embedding are **bit-identical** either way (we verified this). The 60+ **padding positions**, however, change, and the U-Net cross-attends to all 77. So SD v1.5 has learned to read conditioning from the padding tokens, which is a useful fact for later experiments that edit embeddings. Cached pooled embeddings and the checks in Section 5 are unaffected.

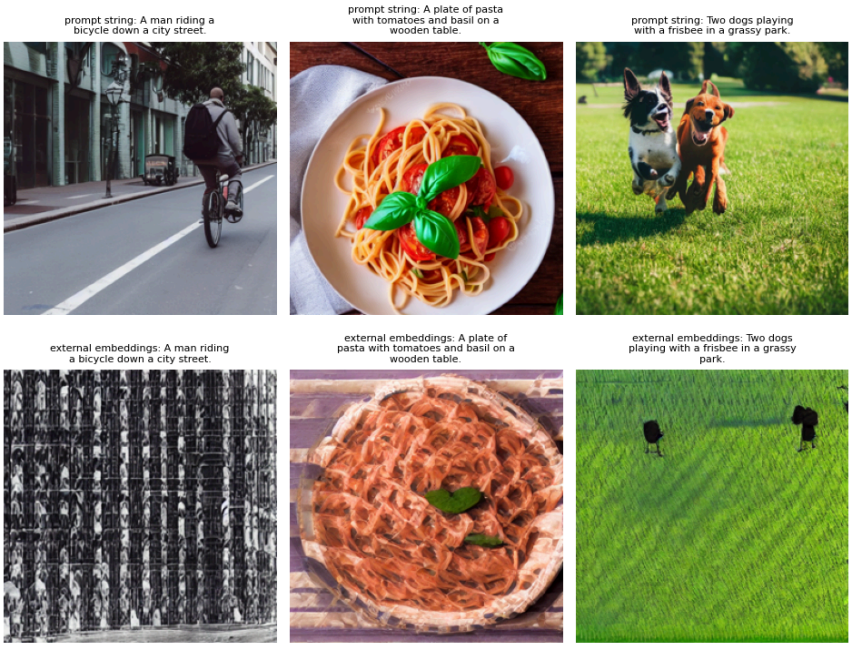


Figure 6. *Conditioning check, first run: SD's built-in prompt encoding (top) vs. our encoder with an attention mask (bottom), first 3 of the 5 prompts. Masking changes the padding tokens and degrades the images.*

After the fix (no mask, matching SD): the 77×768 tensors from our encoder equal the pipeline's own text encoding exactly (max abs difference 0.0 at every position), and all 5 images are pixel-identical (mean abs difference 0.0 / 255). External embeddings from `TextEncoder` can therefore drive the U-Net in later milestones with no hidden change in conditioning. A full rerun of steps 01–04 also reproduced every other result and sample image byte for byte.

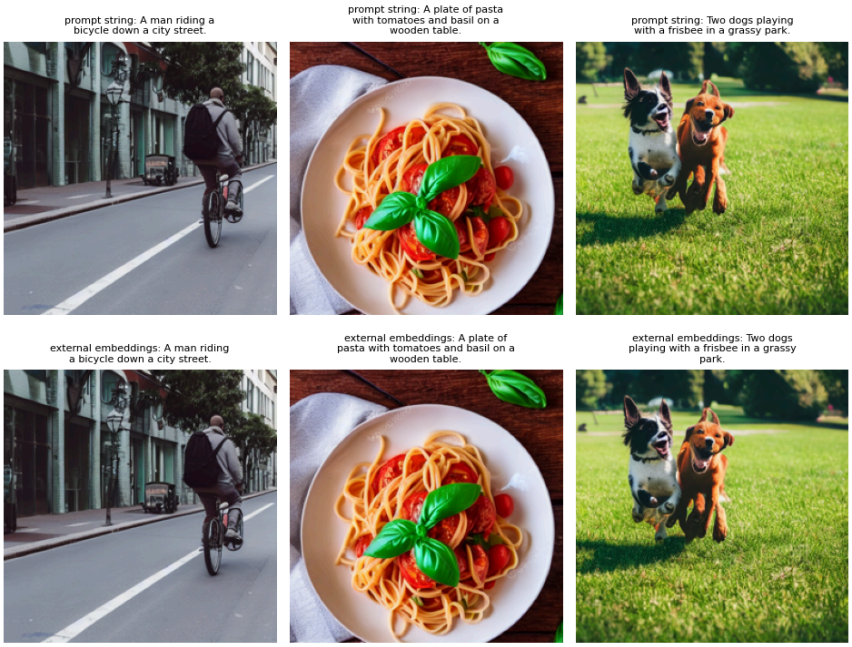


Figure 7. *Conditioning check after the fix: SD's built-in prompt encoding (top) and our encoder without the mask (bottom) give pixel-identical images.*

7. Plan

Milestone	Week	Work
1	1–2	dataset, environment, text→embedding pipeline, 5 test samples, repo (this report)
2	3–4	baseline generation over COCO captions; guidance scale $w \in \{1, 3, 5, 7.5, 10, 15\}$; schedulers × steps {10, 25, 50}
3	5	FID / IS (torchmetrics) on 1–2k images vs. COCO val references; CLIP score; sensitivity plots; text-encoder comparison
4	7 (Dec 13)	final report, gallery, slides and demo video

Extensions if time allows: GPT-style text embeddings, style control through extra conditioning, a web demo.

8. Roles

Member	Responsibilities
Anshuman Singh	model integration, diffusion image generation, repository management
Rohan Prakash Krishna Prakash	dataset collection, cleaning, preprocessing and validation
Sree Ramya Pasala	CLIP embedding pipeline, evaluation, visualizations, technical documentation

9. Ethical considerations (planned)

COCO's demographic and geographic skew; social stereotypes inherited from SD's LAION training data; misuse (deepfakes, non-consensual imagery); and copyright and artist-style imitation. The safety checker is disabled for benign COCO prompts (false positives black out images and bias FID) and enabled for any public demo.